

PROFESSIONAL INSTALLATION MANUAL

Enterprise Server Deployment and Operating System Installation

ITEP 414 – System Administration and Maintenance

Week 3 Portfolio Project

Bachelor of Science in Information Technology (BSIT)

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Ubuntu Server 24.04.4 LTS • VirtualBox • SSH • Server Verification

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1. Introduction

This professional installation manual documents the deployment of an Ubuntu Server virtual machine for a simulated startup enterprise environment. The installation was completed using Ubuntu Server 24.04.4 LTS in Oracle VirtualBox. The server was configured with the required hostname, administrative user account, storage, networking, and OpenSSH Server, then verified using standard Linux commands.

The purpose of this manual is to provide a reproducible procedure that another administrator can follow when deploying a similar Ubuntu Server virtual machine.

2. Objectives

The deployment objectives are to install Ubuntu Server in a virtual machine, configure essential server settings, enable secure remote administration using SSH, verify server functionality, understand the Ubuntu boot process, and document the deployment professionally.

3. Software and Virtual Machine Requirements

Software used:

- Oracle VirtualBox
- Ubuntu Server 24.04.4 LTS ISO

Component	Configuration
VM Name	Ubuntu-Server-Week03
Operating System	Ubuntu Server 24.04.4 LTS
RAM	4 GB
CPU	2 virtual processors
Storage	40 GB VDI
Network	NAT / DHCP
Hostname	server01
User	regie (non-root administrative user)
SSH	OpenSSH Server

4. Ubuntu Server Installation Guide

4.1 Start the Virtual Machine

Create or select the Ubuntu-Server-Week03 virtual machine in VirtualBox and attach the Ubuntu Server 24.04.4 LTS ISO as the optical installation media.

4.2 Language and Keyboard

Select English and configure the appropriate keyboard layout.

4.3 Network Configuration

Accept DHCP unless a static configuration is specifically required. Record the IPv4 address assigned to the server for later verification.

4.4 Hostname and User Account

Configure the hostname as **server01**. Create a non-root administrative user and assign a strong password.

4.5 Storage

Use Guided – Use Entire Disk for the 40 GB virtual disk. The deployment used the full virtual disk and did not enable LVM encryption.

4.6 SSH Server

Enable **Install OpenSSH Server** during installation. This provides secure remote administration capability.

4.7 Additional Packages

Do not install additional snaps unless instructed. The deployment did not select additional packages.

4.8 Complete Installation

After installation completes, remove or detach the Ubuntu installation ISO before rebooting so the VM boots from the installed virtual disk.

5. Initial Server Configuration

After logging in, verify the configured hostname, network address, internet connectivity, and SSH service. The server was also updated using the required APT commands.

System update commands:

```
sudo apt update
sudo apt upgrade -y
```

The update process completed and returned to the normal shell prompt.

6. Server Verification Procedures

6.1 Verify Hostname

Run:

```
hostname
```

Expected/observed result: **server01**.

6.2 Verify IP Address

Run:

```
ip addr
```

Verify that the server has a DHCP-assigned IPv4 address.

6.3 Test Internet Connectivity

Run:

```
ping -c 4 google.com
```

Successful replies with 0% packet loss indicate working connectivity.

6.4 Verify SSH Service

Run:

```
systemctl status ssh
```

The SSH service should display **active (running)**.

7. Troubleshooting Guide

7.1 Ubuntu installer remained at “Reading package lists”

The archive mirror was adjusted and the installation was allowed to continue.

7.2 VM returned to the installer after completion

The Ubuntu ISO was removed from the virtual optical drive before rebooting.

7.3 Hostname did not initially match the module

The hostname was changed to **server01** using `hostnamectl`.

7.4 SSH status needed verification

The SSH service was restarted and verified until it displayed **active (running)**.

8. Best Practices

- Use the required VM specifications and keep installation media separate from the installed virtual disk.
- Use a non-root administrative account and protect it with a strong password.
- Enable SSH only when remote administration is required and verify its service status.
- Update the server after installation using the system package manager.
- Verify hostname, IP address, connectivity, and essential services after deployment.
- Capture clear screenshots as evidence for each major verification step.
- Document installation problems and their solutions so future administrators can reproduce the deployment.

9. Windows Server Evaluation Summary

A separate Windows Server Evaluation virtual machine was installed as the Week 3 bring-home activity. An Administrator password was configured, the system was successfully accessed, and the computer name was configured as **WIN-SERVER-WEEK03**. Screenshots were captured as installation evidence.

10. References

- Ubuntu Server Documentation — <https://documentation.ubuntu.com/server/>
- UEFI Forum — <https://uefi.org/>
- Microsoft Learn — <https://learn.microsoft.com/>
- Rocky Linux Documentation — <https://docs.rockylinux.org/>